



Bayesian Classification

Naive Bayes Classifier

- Naive Bayes is a probabilistic machine learning algorithm that can be used in a wide variety of classification tasks.
- Bayesian classifier can **predict the probability** that a given tuple belongs to a particular class
- Naive Bayes classifiers are based **on Bayes' Theorem**.
- Bayesian classifier **gives more speed and accuracy** as compared to Decision tree classifier.

Bayes' theorem

➤ Bayes' theorem provides a way to calculate the **probability of an event, based on prior knowledge related to that event.**

➤ It's expressed as:

$$P(A|B) = P(B|A) * P(A) / P(B)$$

➤ $P(A|B)$ is the **posterior probability** (Probability of hypothesis A on the observed event B.)

➤ $P(B|A)$ is the **likelihood probability** (Probability of the evidence given that the probability of a hypothesis is true)

➤ $P(A)$ and $P(B)$ are the probabilities of events A and B.

Finding the probability of having a disease given you were tested positive

Finding the probability of liking Harry Potter given we know the person likes fiction

Why is it called 'Naive'?

- The name naive is used because it assumes the features that go into the model is independent of each other.
- That is changing the value of one feature, does not directly influence or change the value of any of the other features used in the algorithm.

Naive Bayes Classifiers

- The dataset is divided into two parts, namely
 - Feature matrix
 - Response vector
- **Feature matrix** contains all the vectors (columns) of dataset in which each vector consists of the value of **dependent features**.
- **Response vector** contains the **value of class variable for each row of feature matrix**.

Working of Naive Bayes Classifiers

- Convert the given dataset into **frequency tables**.
- **Generate Likelihood table** by finding the probabilities of given features.
- Now, use Bayes theorem to **calculate the posterior probability**.

	Outlook	Play
0	Rainy	Yes
1	Sunny	Yes
2	Overcast	Yes
3	Overcast	Yes
4	Sunny	No
5	Rainy	Yes
6	Sunny	Yes
7	Overcast	Yes
8	Rainy	No
9	Sunny	No
10	Sunny	Yes
11	Rainy	No
12	Overcast	Yes
13	Overcast	Yes

Advantages of Naive Bayes Classifiers

- Naïve Bayes is one of **the fast and easy ML algorithms** to predict a class of datasets.
- It can be **used for Binary as well as Multi-class Classifications**.
- It performs well in Multi-class predictions as compared to the other Algorithms.
- It is the most popular choice for text classification problems.

Disadvantages of Naive Bayes Classifiers

- Naive Bayes assumes that all features are independent or unrelated, so it cannot learn the relationship between features.

Applications of Naive Bayes Classifiers

- It is used for Credit Scoring.
- It is used in medical data classification.
- It can be used in real-time predictions because Naïve Bayes Classifier is an eager learner.
- It is used in Text classification such as Spam filtering and Sentiment analysis.